

JanSetu AI (■■■■■ AI)

Digital Public Infrastructure for Multilingual Citizen Voice & Predictive Municipal Governance

Dual-Cadre Governance Seamlessly bridges grassroots citizens and municipal administrators through dedicated, cadre-isolated portals.	Zero-Barrier Voice AI Sarvam AI Indic speech-to-text + Gemini 2.0 enabling voice grievances in Hindi, English, and regional dialects.
Predictive PostGIS GIS Transforms reactive complaint-handling into spatial distress hotspot clustering & proactive civic works planning.	Anti-Hallucination Policy RAG pgvector semantic retrieval grounding development proposals directly in AMRUT 2.0, JJM, and PMGSY guidelines.

The Structural Disconnect in Urban Governance

Why existing grievance portals (CPGRAMS/CM Helpline) fail grassroots citizens and administrators

Linguistic & Literacy Exclusion 65%+ of rural & semi-urban citizens cannot navigate complex English/formal Hindi web forms. Dialects (Bundelkhandi, Bagheli) are ignored.	Siloed, Disconnected Data Public works budgets, Census demographics, and engineering asset registers exist in isolated silos with zero spatial correlation.
Reactive Firefighting Municipal bodies fix complaints individually rather than identifying recurring spatial clusters (e.g., repeating pipe bursts or water contamination).	Subjective Capital Allocation Infrastructure projects are often sanctioned based on political lobbying or subjective discretion rather than objective distress metrics.

DPI-First, Cadre-Isolated Governance Engine

Modular Digital Public Good built on open standards, PostGIS, and sovereign Indian AI models

Layer 1: Citizen Voice Intake Multimodal filing via conversational voice notes, bilingual text, and geocoded photo uploads with instant SLA tracking.	Layer 2: AI Triage & NLP Pipeline FastAPI microservice executing Sarvam AI ASR, Google Gemini 2.0 category extraction, sentiment, and urgency scoring.
Layer 3: Spatial GIS & Fusion NestJS API Gateway fusing PostGIS coordinates with Census demographics, ward GIS polygons, and PHE asset registers.	Layer 4: Decision Cockpit & DPR Interactive Leaflet GIS map, 5-factor priority formula, and pgvector RAG synthesizing audit-ready development proposals.

Conversational, Multilingual Voice-First Intake

Ensuring no Indian citizen is left behind due to language, dialect, or literacy barriers

Sarvam AI Indic Speech-to-Text Trained on Indian regional accents and acoustic conditions, accurately transcribing spoken audio into structured text.	Dialect AI Chatbot & Form Interactive AI assistant capable of conversational interview, asking clarifying questions, and auto-filling municipal categories.
Automated Geocoding & Photo Capture HTML5 geolocation captures precise GPS coordinates with street address reverse-geocoding and image verification.	Real-Time SLA & Status Tracking Every ticket generates an audit-verifiable tracking code (e.g., GWL-2026-4821) with milestone resolution progress.

Spatial Data Fusion Across Siloed Datasets

Fusing citizen demand with Census demographics and public engineering infrastructure registers

Gwalior GIS Ward Boundaries Accurate GeoJSON polygon layers representing all 66 municipal wards of Gwalior Municipal Corporation (Morar, Lashkar, Gwalior Fort).	Demographic & Vulnerability Layer Ward-level Census population density, SC/ST demographic percentages, and economic vulnerability indicators.
Infrastructure Asset Registers Overlaid PHE water pipeline age, road maintenance logs, storm drain capacities, and street lighting distribution.	Spatial Lineage & Auditability Every data point tracks source agency, ingestion timestamp, and confidence score for complete administrative transparency.

Automated Geospatial Distress Clustering

Moving governance from reactive single complaints to predictive community infrastructure repair

PostGIS Spatial Clustering (ST_DWithin) Clusters complaints occurring within a 300m radius and 14-day temporal window into unified spatial distress hotspots.	Root Cause Detection Identifies when 15 individual drinking water complaints in Ward 22 actually stem from a single cracked main distribution pipe.
Hotspot Severity Index Combines complaint velocity, affected population radius, and public health risk (e.g. sewage mixing with drinking water).	Automated Work Order Triggers Transfers cluster reports directly to executive engineers with auto-calculated resource and equipment requirements.

Deterministic 5-Factor Priority Scoring Engine

Eliminating subjective bias through a transparent, audit-verifiable mathematical formula

1. Demand Intensity (30%) Normalized volume and frequency of complaints from verified local residents within the cluster zone.	2. Infrastructure Gap (25%) Physical age and degradation rating of public assets (e.g., pipelines over 20 years old, unpaved drainage).
3. Vulnerable Population Impact (20%) Proportion of affected low-income, marginalized, and high-density communities benefiting from the intervention.	4. Historical Deficit (15%) & Budget (10%) Historical municipal spending neglect weighted against capital expenditure feasibility under current financial cycles.

Anti-Hallucination Policy RAG & DPR Generator

Matching local ward distress with central/state welfare schemes to draft audit-ready project reports

pgvector Knowledge Base Ingested official operational guidelines: AMRUT 2.0, Jal Jeevan Mission, PMGSY, Swachh Bharat Mission Urban.	Cosine Similarity Semantic Search Retrieves exact funding clauses, matching grant formulas (60% Central / 40% State), and technical specifications.
Auto-Generated DPR Proposals Drafts Detailed Project Reports (DPR) with estimated CAPEX, target ward, beneficiary count, and scheme eligibility citations.	Zero-Hallucination Guarantee Prompts strictly constrain AI responses to verbatim retrieved policy documents, preventing fabricated schemes or budgets.

High-Performance 3-Tier Microservices

Cloud-native, horizontally scalable architecture ready for deployment on national infrastructure

Frontend (React 18 + Vite) Tailwind CSS, Leaflet GIS, responsive mobile-first UI, client-side caching, and sub-second page transitions.	API Gateway (NestJS + TypeScript) Modular controllers, strict DTO validation pipes, Swagger OpenAPI documentation, and role-based guards.
AI Microservice (FastAPI + Python) Asynchronous FastAPI runtime orchestrating Google Gemini 2.0 and Sarvam AI Indic speech pipelines.	Database & Vector (Supabase PostGIS) PostgreSQL 15 with PostGIS spatial indexes (GIST), pgvector (vector(768)), and connection pooling.

Cadre Isolation & Zero-Trust Citizen Privacy

Enterprise role-based access control protecting citizen PII while enabling transparent policymaking

Cadre Isolation Architecture Strict boundary separating Citizen and Policymaker views. Citizens cannot see officer notes; officers cannot alter citizen filings.	Citizen PII Anonymization Policymaker GIS heatmaps display aggregated demand points without names, phone numbers, or private identity markers.
Supabase JWT & Google OAuth 2.0 Cryptographically verified authentication tokens with automatic profile synchronization and self-healing sessions.	Immutable Audit Trail Every status update, scheme recommendation, and priority score computation is logged with actor timestamp.

Gwalior Municipal Pilot Case Study

Demonstrated operational success in Ward 22 Morar with measurable civic metrics

Pilot Zone: Ward 22 Morar High-density residential and commercial zone historically facing severe water pressure drop and pipeline contamination.	Hotspot Detection in <48 Hours JanSetu AI clustered 28 voice and text grievances into a single 250m pipeline breach hotspot near Morar Subhash Chowk.
64% Faster Resolution Cycle Average grievance resolution dropped from 14.2 days down to 5.1 days through automated cross-departmental dispatch.	AMRUT 2.0 Scheme Linking Automatically drafted a Rs 42.5 Lakh pipeline rejuvenation DPR backed by AMRUT 2.0 Sub-Mission for Water Supply.

National Scale: Ready for 4,000+ Urban Local Bodies

Built as open Digital Public Infrastructure for pan-India municipal integration

Pan-India ULB Replicability Zero municipal-specific hardcoding. Ingesting any new city's GeoJSON boundaries deploys full functionality in minutes.	India Stack & Bhashini Integration Pre-architected for DigiLocker document attachment, Aadhaar e-KYC validation, and national Bhashini voice pipelines.
Open API & Digital Public Good (DPG) Complete OpenAPI Swagger specifications enabling third-party civic tech apps, research bodies, and state portals to connect.	Sustainable Civic ROI Reduces municipal operational overhead, prevents catastrophic infrastructure failures, and fosters citizen trust in governance.